

The Social Mechanics of Market Anomalies: A Machine Learning Analysis of Reddit's Influence on Meme Stocks

MinSeo Kim
2025.06.15

1. Motivation and Problem Definition

The "meme stock" phenomenon in early 2021, and in particular the GameStop (GME) incident, constitutes a significant distortion of historical financial market trends witnessed recently. During this time, dramatic price surges in stocks occurred, which were not fueled by traditional fundamental analysis but rather by collective action among retail investors, largely organized over internet social media platforms (Aggarwal et al., 2024; Costola et al., 2025). The events contradicted the dominant principles of market efficiency and underscored the increasing power of behavioral finance and internet communities on asset pricing (Shiller, 2000; Barberis & Thaler, 2003).

Meme stocks are, by definition, marked by price action that is determined more by social media activity than by the underlying financial health of the organization. The outcome is intense volatility and unusually heavy trading volumes fueled chiefly by individual, as opposed to institutional, participants (Kalikan, 2021; Monmouth University, 2023). The GameStop incident demonstrated the growing influence of retail traders in the markets of today, a trend facilitated by user-friendly trading software and reinforced by internet coordination.

Although short squeezes have always been a feature of market movement, the one in GameStop exceeded conventional description because of their unprecedented magnitude and novel catalysts (Monmouth University, 2023). In a normal short squeeze, highly shorted stocks undergo swift price appreciation that compels short sellers to cover shorts to limit losses, thus intensifying the price rally even more. But the GameStop incident was characterized by a never-before-seen degree of coordination among retail investors enabled by social media platforms.

Retail traders, specifically those mobilizing on Reddit communities, spotted highly shorted shares such as GameStop and collectively engaged in purchases that sought to defy institutional short sellers. This collective action, augmented by online forums and viral information, unleashed historic short squeezing that imposed massive monetary losses on institutional investors (Monmouth University, 2023). The event demonstrates the power of social media in facilitating collective actions among individual traders, thus exerting a profound impact on price movements (Costola et al., 2025).

Building from the context above, this study is poised to carry out an empirical examination of the correlation between Reddit forum activities and the notable market behaviors noticed in GameStop. The primary aim is to empirically determine the contribution of Reddit forums, r/wallstreetbets, and subreddits dedicated to GME, to these market irregularities and their potential in the future to predict daily stock price movements in GME.

This analysis is utilized as the foundation to develop a conceptual internet-based community tracking model. The model would be capable of leveraging the output of this research to potentially detect early indications of similar social media-driven market events, providing valuable insights for retail investors, institutional participants, and market regulators (Costola et al., 2025).

2. Dataset Description and Preprocessing

Our study relies on a wide range of historical stock information and Reddit social media news spanning the pivotal time of the meme stock trend.

2.1 Dataset Description

The following datasets form the foundation of the study:

- GameStop (GME) Stock Data (dedicated to the public domain): A large collection of historical trading volume and price data for GME from February 13, 2002, to May 13, 2024. The data consists largely of daily entries for Date, Open (opening price), Close (closing price), and Volume.
- Reddit WallStreetBets Posts (dedicated to the public domain): Large dataset consisting of more than 32 million comments. Some of the main features in this raw file were author (comment author), score (upvotes on the comment), body (comment text content), and created_utc (Unix creation timestamp). This raw file needed substantial tuning because of the size and intrinsic data quality issues.
- Reddit WSB Comment Data (dedicated to the public domain): Contains four datasets of more niche subreddits: r/GME (January 4, 2021, to December 31, 2021), r/DDintoGME (April 7, 2021, to January 31, 2022), r/GMEJungle (July 14, 2021, to January 31, 2022), and r/superstonk (March 15, 2021, to February 14, 2022). Each of

the four data sets contains id, title, score, author, num_comments, date (human-readable timestamp), and valuable pre-computed sentiment scores (compound, neg, neu, pos), along with Natural Language Processing (NLP) features such as clean_text.

- **Reddit Dataset on Meme Stock GameStop (Han, 2022):** Contains four datasets from more focused subreddits: r/GME (January 4, 2021, to December 31, 2021), r/DDintoGME (April 7, 2021, to January 31, 2022), r/GMEJungle (July 14, 2021, to January 31, 2022), and r/superstonk (March 15, 2021, to February 14, 2022). All four datasets include id, title, score, author, num_comments, date (human-readable timestamp), and crucial pre-computed sentiment scores (compound, neg, neu, pos), along with Natural Language Processing (NLP) features such as clean_text.

2.2 Data Preprocessing

To enable this analysis, the data was limited to the core period of the meme stock phenomenon.

(1) GME Stock Data Preparation:

Once the file was loaded, and its 'Date' column was specifically converted into datetime objects to enable time-series operations. Data was resampled into daily frequency and missing values (weekends, market holidays) of 'Open', 'Close', and 'Volume' were replaced with a forward-fill strategy for data consistency. Stock data was then precisely filtered to align with the main Reddit activity window, i.e., 09.29.2020 - 02.14.2022. This ensured stock data precisely matched the corresponding period of available social media data for proper comparison.

(2) Reddit Comment Data Refinement:

Because of the sheer size of the raw comment file (more than 11GB) and patterns of data corruption observed outside of roughly 30 million rows, only the first 30 million rows were processed to preserve data integrity and computational realities. The 'score' field was converted strictly to numeric type, converting any non-numeric values to 'NaN' and then handled accordingly. The 'created_utc' column, with Unix timestamps, was strongly converted to datetime objects, in this case using the 'unit='s' parameter and coercing errors to handle potential inconsistencies. Rows flagged as having missing comment text ('body'), invalid timestamps, or invalid numerical scores (after conversion) were systematically removed to remove faulty entries.

These cleaned comments were then temporally filtered to the 09.29.2020 to 02.14.2022 time frame, yielding approximately 2 million clean and contextually relevant comments for analysis. VADER (Valence

Aware Dictionary and sEntiment Reasoner) sentiment analysis method was used on the cleaned text comments to produce a 'compound_sentiment' score for each comment, giving a quantitative measure of emotional tone conveyed.

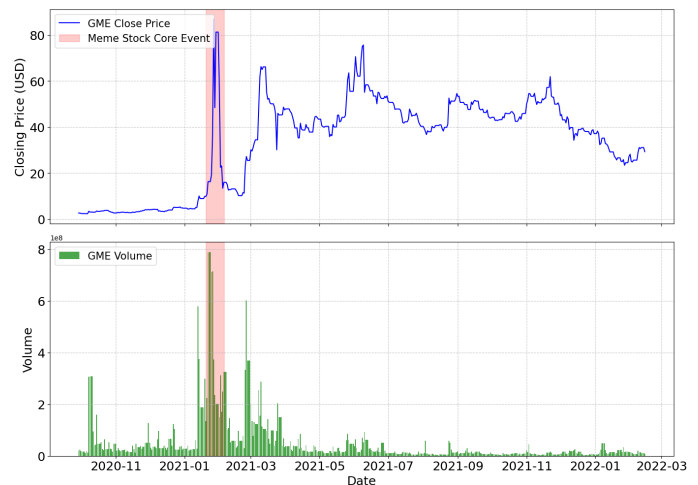


Fig. 1. Daily GME Stock and Volume until 05.13.2024



Fig. 2. Daily GME Stock and Volume during core Reddit activity period (09.29.2020 to 02.14.2022)

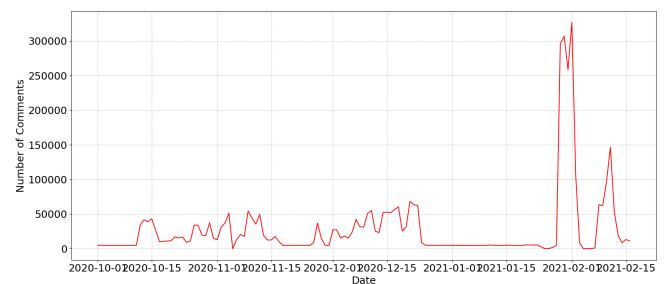


Fig. 3. Daily number of comments in subreddit r/wallstreetbets

(3) Aggregation of Reddit Post Data:

All five Reddit post datasets (r/wsb, r/GME, r/DDintoGME, r/GMEJungle, r/supertonk) were loaded and merged into one unified DataFrame. Column names in various datasets (such as 'date', 'created') were unified to 'timestamp_utc'. Data types of 'score' and 'num_comments' were expressly defined as numeric. For subreddits without pre-calculated

sentiment scores (e.g., r/wallstreetbets), VADER sentiment analysis was directly computed on their 'clean_text' (or computed on 'title' where 'body' text wasn't available), providing the complete sentiment profile for all the posts. The merged posts were then filtered to the overall analysis timeframe of 09.20.2020 to 02.14.2022, encompassing all post activity of interest.

(4) Daily Metric Aggregation:

The merged and filtered Reddit posts were then extracted at a daily frequency to get subreddit-specific activity and sentiment. This involved computing the daily total post frequencies and the daily average compound sentiment for each of the five subreddits. The dates with zero activity for a specific subreddit were filled accordingly with a value of 0.

(5) Feature Engineering for Machine Learning:

The ultimate target variable for the exercise in regression was set as 'Target_GME_Next_Day_Pct_Change', the next day's percentage change in the 'Close' price of GME on the next trading day. For exploratory reasons, 'Target_GME_Next_Day_Open_Pct_Change' was also created. To capture potential delayed impacts, lagged versions of all of the daily post quantities and mean sentiments (for all five subreddits) were created with 1-day and 3-day lags. This is in response to the reality that social media activity today will not necessarily impact stock prices for days after today. To reduce noise and extract underlying volatility or trends in social media activity, 3-day and 7-day rolling means and standard deviations were calculated for all daily post volumes and average sentiments. These rolling features were also shifted by one day to prevent data leakage from the future. The GME stock data and the complete set of engineered Reddit features were then merged into a DataFrame (referred to as 'X'), using their common date index. Rows containing any lingering 'NaN' values, mostly due to the rolling and shifting operations, were methodically removed in order to be left with a tidy dataset with which to train models. The final 'X' DataFrame was GME's own daily metrics ('Open', 'Close', 'Volume', 'GME_Close_Pct_Change', 'GME_Open_Pct_Change') joined with all of the engineered lagged and rolled Reddit features (yielding 75 distinct features). The 'y' DataFrame was 'Target_GME_Next_Day_Pct_Change'.

(6) Data Splitting:

The prepared dataset was subsequently divided into 80% training and 20% testing sets. The division was performed along the time dimension, thus the models were trained on past data only and tested on future,

unknown market time periods, which is crucial in order to make solid time-series predictions.

3. Machine Learning Algorithms

To tackle the regression task of predicting GME's next-day percentage change, five different machine learning algorithms were chosen and applied. This varied choice — ranging from simple linear models and robust ensemble methods to a neural network and a complex non-linear algorithm — was intended to delve into the intricate relationships within the data and determine the most appropriate method for this difficult prediction task.

1. *Linear Regression*: Linear Regression is a simple supervised learning algorithm for predicting a continuous dependent variable (say, 'Target_GME_Next_Day_Pct_Change') from one or more independent variables (our features). It achieves this by fitting a linear equation to observed data in such a way that the sum of the squared differences between predicted and actual values is minimized. Selected as an important benchmark, simplicity and interpretability allow for the initial evaluation of linear relationships and benchmark that more sophisticated models can be compared to.
2. *Random Forest Regressor*: Random Forest Regressor is an ensemble learning technique that takes the predictions of numerous decision trees and averages them. The trees are trained separately on a bootstrapped sample of the training data and a random selection of features. For regression, the final prediction is the average of the predictions made by every tree. This algorithm was chosen due to its increased accuracy, innate resistance to overfitting, and capacity to deal with non-linear relationships and interactions within complex datasets effectively.
3. *Neural Network (MLP Regressor)*: A Multi-layer Perceptron (MLP) Regressor is a more sophisticated non-linear modeling technique, inspired by the human brain. An Artificial Neural Network (ANN) has interconnected layers of nodes (neurons) that process information. MLP Regressors can learn complicated non-linear relationships between inputs and outputs and thus are appropriate for intricate patterns in data where linear models may not be the best. It was added to see whether a more abstract, deep learning-style architecture would be capable of finding predictive signals.
4. *XGBoost Regressor*: XGBoost (Extreme Gradient Boosting) Regressor is a distributed, speed-optimized, high-performance gradient boosting library. It is an ensemble approach that generates

decision trees sequentially, with each additional tree correcting the mistakes of the former, and in doing so, gradually enhancing the precision of the model. It is renowned for its scalability, flexibility, and performance and has a tendency to deliver state-of-the-art outcomes on structured data prediction problems. It was used to take advantage of the strength of gradient boosting for robust regression.

5. *Support Vector Regressor (SVR)*: Support Vector Regressor (SVR) is a form of Support Vector Machines (SVMs) for regression tasks. In contrast to conventional regression models that attempt to minimize error, SVR attempts to determine a hyperplane that maximally fits the data points in a defined margin (ϵ), accepting errors within this margin and penalizing larger deviations. SVR is effective in high-dimensional space and for the modeling of non-linear relationships through different kernel functions (e.g., Radial Basis Function - RBF kernel, polynomial kernel).

4. Model Training and Performance Evaluation

The data prepared, including engineered features (X) and the target variable (y), was trained and evaluated rigorously with the chosen machine learning models. The main intention was to evaluate their effectiveness in predicting tomorrow's GME percentage price change.

4.1. Data Splitting and Feature Scaling

Before training the models, the data was divided into training and test sets to allow for orderly evaluation. Since the data included a time-series feature, the data was divided along time to avoid data leakage from future observations. In particular, the first 80% of data in chronological terms was used for training and the remaining 20% for testing.

This gave us a training set of 396 samples (October 6, 2020, to November 5, 2021) and a test set of 100 samples (November 6, 2021, to February 13, 2022). Feature scaling was then performed using 'StandardScaler'. This preprocessing step normalizes the features to be mean 0 and standard deviation 1. Scaling is especially important for feature magnitude sensitive algorithms like Linear Regression, Neural Networks, and Support Vector Regressors to enable fast convergence and avoid larger scale features overpowering the learning process. The scaler was trained solely on the training data and then applied to both the training set and test set, taking great care not to leak data.

4.2. Evaluation Metrics

Performance of models on this regression problem was evaluated quantitatively with three common metrics:

- Mean Squared Error (MSE): Mean of the squared errors (mean squared difference between forecasted and observed values). The smaller the MSE, the better the model.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Root Mean Squared Error (RMSE): Square root of MSE, which gives an error measure in the same unit as the target variable (percentage change) and thus is more interpretable. Lower RMSE means good model fit.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- R-squared (R^2): The percentage of the variance in the dependent variable that is explained by the independent variables. A perfect fit is an R^2 of 1.0, and the model explains 100% of the variance in the target. An R^2 of 0.0 indicates that the model is no better than forecasting the mean of the target variable. Most of all, a negative R^2 means the model's predictions are worse than predicting the mean, which suggests that the model is not picking up on any helpful patterns and its error is greater than the variance of the data itself.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

4.3. Model Performance Results

All chosen models were trained on the scaled training set (or unscaled for Random Forest since it is less sensitive) and then tested on the unseen test set. Performance metrics are shown in Table 1.

4.4. Analysis of Performance

The performance measures always mirrored the profound difficulty in forecasting daily percentage movements in GME's stock price during the given timeframe.

All the models yielded negative R-squared values, meaning that none of them were able to capture predictive patterns that enabled them to beat a naive

baseline of predicting the average next-day price change.

This outcome is not unexpected in the highly volatile and inherently unpredictable arena of short-term financial market forecasting, particularly for instruments that are prone to dynamic and often irrational collective behavior. Of the algorithms tested, the Support Vector Regressor (SVR) was the "most appropriate" model relative to its performance with an R2 score of -0.0193. Although it is still negative, it is nearer to 0 compared to the other models (e.g., Linear Regression = -1.6551, MLP = -3.7557, XGBoost = -0.9197, and Random Forest = -0.6379).

This implies that SVR, for all its general weaknesses here, was best able to fit the data without doing substantially worse than the mean prediction. The Neural Network (MLP) also issued a 'Convergence Warning', that it did not converge within the specified maximum number of iterations, which again hints at the complexity of the underlying data.

The poor performance across model types, despite heavy feature engineering, attests to the underlying noise and non-linear, frequently immeasurable, drivers at play in the meme stock phenomenon. It underscores the argument that although social media certainly can have an impact on market action, it is extremely challenging to translate that impact into accurate short-term price forecasts.

Model	MSE	RMSE	R2
Linear Regression	57.6552	7.5931	-1.6551
Random Forest	35.5666	5.9638	-0.6379
Neural Network	103.2708	10.1622	-3.7557
XGBoost	41.6859	6.4565	-0.9197
SVR	22.1332	4.7046	-0.0193

Table 1. Model performance results

5. Discussion

Empirical examination of the interrelationship between Reddit community activity and GameStop stock price movement in the 2021 meme stock event has provided some intriguing findings.

5.1. Outcomes and Model Performance

The primary objective of developing a predictive model of GME's next-day percentage price change was very challenging.

Despite exhaustive preprocessing and significant feature engineering (including lagged and rolling daily post counts and average sentiment of five key subreddits), all five machine learning models attempted (Linear Regression, Random Forest Regressor, Neural Network, XGBoost, and Support Vector Regressor) yielded negative R-squared values on the unseen test data. This across-the-board poor performance indicates that none of the models successfully learned to predict patterns to outperform a naive benchmark of predicting the mean. The Support Vector Regressor (SVR) had the "best" comparative performance with an R2 of -0.0193, which implies it performed most like this benchmark. This result highlights the intrinsic challenge of predicting short-term price movement of extremely volatile instruments such as meme stocks, where fast information dissemination and non-fundamental factors tend to dominate recognizable patterns.

5.2. Reddit's Role: Insights from Exploratory Analysis

Notwithstanding the difficulty in direct price prediction, our exploratory data analysis yielded ample proof of Reddit's central position in the GME drama.

Plots of the number of posts per day by subreddits illustrated massive, concurrent spikes in online discourse during the times of GME's extreme volatility (Figure 4, 5). Subreddits such as r/wallstreetbets and r/superstonk saw hundreds of thousands of posts per day, obviously pointing to focused retail investor interest. The different start times and growth patterns of the subreddits depicted the active expansion of the community.

Notably, some negative correlations with trading volume were observed, which suggested complex behaviors like the "HODL" (Hold On for Dear Life) mentality, where extended positive sentiment would counterintuitively lead to lower trading volume as investors wait out selling.

5.3. Proposed Online Community Monitoring Model

Based on the experience of Reddit behavior during the GME event, we suggest a conceptual online community monitoring model. The system is to be utilized as an early warning or information system for market participants by detecting abnormal social media trends.

The model would monitor close to real-time online community statistics (e.g., Reddit) to identify statistically significant outliers in posting frequency, sentiment changes, and keyword frequencies.

The key functionalities would be a Data Ingestion Module, an Advanced NLP and Feature Extraction Module (topic modeling and sentiment analysis), an

Anomaly Detection Engine (unsupervised ML or statistical algorithms), and an Alerting and Visualization Dashboard. This model would have practical implications for institutional traders, retail investors, and regulators.

5.4. Limitations

There were several limitations that this research faced. The very nature of financial market volatility, especially those social media-influenced markets, is a primary limitation, as reflected in our models' negative R2 values. The study was confined to Reddit alone and did not include signals from other sites (e.g., X/ Twitter). It is still difficult to establish definitive causation from Reddit activity to the movement of individual stock prices. Additionally, although sophisticated, the numerical features created might be incapable of encapsulating the richness of text nuance, and the regression models used are really only well-suited to linear relationships, whereas meme stock activity is probably highly complicated and non-linear. Data sparsity in early times for later-started subreddits was also a problem. Fig. 4. GME closing price vs. Reddit daily posting volume Fig. 5. GME close price to sentiment in Reddit posts

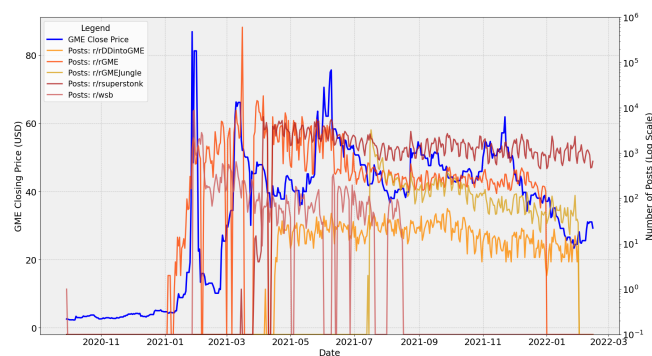


Fig. 4. GME close price to Reddit daily post volume

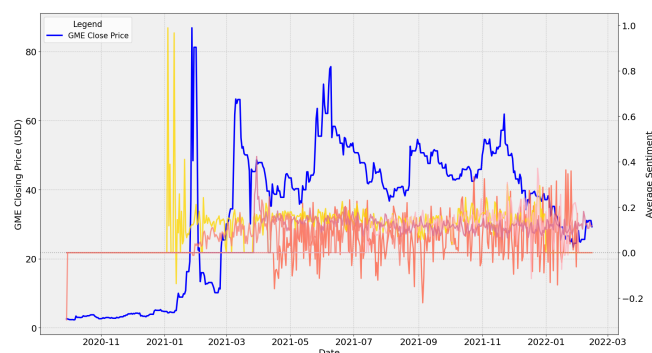


Fig. 5. GME close price to sentiment in Reddit posts

5.5. Future Work

To enhance the predictability and utility of understanding social media's impact on financial markets, several directions are proposed for future work. Future work can move from trying to predict percentage change (regression) to a binary or multi-

class classification problem (e.g., prediction of "up," "down," or "neutral" movement), which will be more tractable results. Sophisticated NLP and Deep Learning methodologies (e.g., transformer-based sentiment, RNN/LSTM for time-series patterns) may be able to detect more intricate relationships. A richer feature space can also include exogenous variables such as conventional market measures (VIX), news sentiment, and company-specific information. Network analysis can look at community structure and information flow. Finally, dynamic feature selection/weighting and reinforcement learning can be explored for adaptive strategies.

References

- [1] Aggarwal, D., Choi, A. H., & Lee, Y.-H. A. (2024). *The meme stock frenzy: Origins and implications*. Southern California Law Review, 97(4), 1387-1450.
- [2] Barberis, N., & Thaler, R. H. (2003). A survey of behavioral finance. In G. M. Constantinides, M. Harris, & R. M. Stulz (Eds.), *Handbook of the economics of finance* (Vol. 1, Part B, pp. 1053-1128). Elsevier.
- [3] Costola, M., Iacopini, E., & Pammolli, F. (2025). A social media alert system for meme stocks. *Journal of Risk Finance*, 26(3), 269-286.
- [4] Kalikan, Z. (2021). *Meme Stocks are here to stay? An empirical analysis on the relationship between investor sentiment and asset performance*. (Master's thesis). Luiss Guido Carli.
- [5] Monmouth University. (2023). *The GameStop Short Squeeze as a Case Study in Business Law Education*. Education Quarterly Reviews.
- [6] Shiller, R. J. (2000). *Irrational exuberance* (1st ed.). Princeton University Press.